

Program : Diploma in Information Technology	
Course Code : 4261	Course Title: Network Switching Technologies
Semester : 4	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- To provide basics of Network Switching Technologies from a Network Administrator's point-of-view.
- To provide the basics of Local Area Network, Network Switching and its Administration.

Course Prerequisites:

Topic	Course Code	Course Title	Semester
Basic knowledge in Network Simulator		Computer Hardware and Networking Lab	3
Subnetting, LAN, IP Address, Switching etc		Fundamentals of Network Administration	3

Course Outcomes:

On completion of the course, students will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Identify how to configure, verify and troubleshoot IPv4 addressing and subnetting for a simple network.	15	Applying
CO2	Identify the basic concepts of Internetwork Operating System and management of Internetwork and Network Devices for a simple Network.	14	Applying
CO3	Identify how to describe, verify, configure troubleshoot switching concepts, port security, VLANs, VLAN Trunking Protocol and Spanning Tree Protocol.	15	Applying
CO4	Identify the basic concepts of Network Security and Network Address Translation for a simple Network.	14	Applying
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Identify how to configure, verify and troubleshoot IPv4 addressing and subnetting for a simple network.		
M1.01	Outline Subnetting and subnet mask.	1	Understanding
M1.02	Summarize Classless Inter-Domain Routing	1	Understanding
M1.03	Identify Class C Address Subnetting for a simple network	3	Applying
M1.04	Identify Class B Address Subnetting for a simple network	3	Applying
M1.05	Identify Class A Address Subnetting for a simple network	3	Applying
M1.06	Summarize Variable Length Subnet Mask Design and Implementation for a simple network.	2	Understanding
M1.07	Outline Troubleshooting IP Addressing	2	Understanding
Contents: Subnetting – Subnetting Steps – Subnet Mask – Classless Inter-Domain Routing – Subnetting Class C Address – Subnetting Class B Address – Subnetting Class A Address Variable Length Subnet Masks – VLSM Design – Implementing VLSM Networks – Troubleshooting IP Addressing			
CO2	Identify the basic concepts of Internetwork Operating System and management of Internetwork and Network Devices for a simple Network.		
M2.01	Summarize Internetwork Operating System (IOS)	1	Understanding
M2.02	Illustrate basic IOS Command-Line Interface (CLI) Commands	2	Understanding
M2.03	Summarize Administrative Configuration IOS	2	Understanding

	CLI Commands		
M2.04	Outline Router and Switch Interfaces	1	Understanding
M2.05	Summarize Viewing, Saving and Erasing Configurations	1	Understanding
M2.06	Illustrate the internal components of a Router and Switch	1	Understanding
M2.07	Identify Backing up and Restoring the Configuration	1	Applying
M2.08	Outline the Configuration of DHCP	1	Understanding
M2.09	Outline usage of Telnet	1	Understanding
M2.10	Outline how to resolve Hostnames	1	Understanding
M2.11	Summarize how to check network connectivity and its troubleshooting	1	Understanding
M2.12	Identify Backing up and restoring IOS	1	Applying
	Series Test – I	1	

Contents:

IOS Basics – Command-Line Interface (CLI) (enable, disable, logout, interface, configure terminal commands) – Administrative Configurations (Hostname, Banners, Passwords Interface Descriptions commands) – Router and Switch Interfaces (Bringing up an interface, configuring an IP address, serial interface command) – Viewing, Saving and Erasing Configurations

Internal components of a router and switch – Backing up and Restoring the Configuration – Configuration of DHCP – Using Telnet – Resolving Hostname – Checking Network Connectivity and Troubleshooting – Backing up and restoring IOS

CO3	Identify how to describe, verify, configure troubleshoot switching concepts, port security, VLANs, VLAN Trunking Protocol and Spanning Tree Protocol.		
M3.01	Summarize Switching Services	2	Understanding
M3.02	Outline Port Security	1	Understanding
M3.03	Identify how to do basic configurations of a Switch	2	Applying
M3.04	Summarize VLAN and how to identify VLAN	1	Understanding
M3.05	Outline Routing between VLANs	1	Understanding
M3.06	Identify how to configure a VLAN	2	Applying
M3.07	Summarize VLAN Trunking Protocol and its modes of operation	1	Understanding

M3.08	Identify how to configure VTP	2	Applying
M3.09	Outline Spanning Tree Protocols with terms, operations and types	2	Understanding
M3.10	Summarize EtherChannel	1	Understanding
Contents: Switching Services – Three switch functions at layer 2 (Address Learning – Forward/Filter decisions, Loop avoidance) – Port Security – Configuring Switches VLAN Basics – Identifying VLANs – Routing between VLANs – Configuring VLANs – Assigning switch ports to VLAN – Configuring Trunk Ports – Defining the allowed VLANs on a Trunk – Changing or Modifying the Trunk Native VLAN. VLAN Trunking Protocol (VTP) – Modes of operation – Configure VTP – Spanning Tree Protocol (STP) – Terms – Operations - Types of STP – EtherChannel			
CO4	Identify the basic concepts of Network Security and Network Address Translation for a simple Network.		
M 4.01	Outline Perimeter Router, Firewall and Internal Router	2	Understanding
M 4.02	Summarize Standard and Extended Access Lists	3	Understanding
M 4.03	Illustrate usage and types of NAT	3	Understanding
M 4.04	Outline Working of NAT	3	Understanding
M 4.05	Identify the configuration of static and dynamic NAT	3	Applying
	Series Test – II	1	
Contents: LAN Security - Perimeter Router, Firewall and Internal Router – Access Lists (Standard and Extended) Network Address Translation (NAT) – Use of NAT – Types of NAT – Working of NAT – Static NAT Configuration – Dynamic NAT Configuration			

Text / Reference:

T/R	Book Title / Author
T1	Todd Lammle, <i>CCNA Routing and Switching Complete Study Guide</i> , 2 nd Edition, Sybex (Wiley) Publication, 2016
R1	Narasimha Karumanchi, Dr. A. Damodaram and Dr. M. Sreenivasa Rao. <i>Elements of Computer Networking: An Integrated Approach</i> . CareerMonk Publications, 2014.

Online Resources:

Sl.No	Website Link
1	https://www.freeccnastudyguide.com/study-guides/icnd1-100-101/
2	https://www.thebryantadvantage.com/videos-and-tutorials/ccna-and-ccent-tutorials/
3	https://www.guru99.com/introduction-ccna.html
4	https://study-ccna.com/
5	http://www.omnisecu.com/cisco-certified-network-associate-ccna/index.php